Is Climate Risk Priced in Mortgage Credit? Evidence from 1.05 Million Freddie Mac Loans

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***Abstract*—Hurricanes make mortgage borrowers miss payments, but do they make lenders lose money? We combine Freddie Mac loan-level data (2004–2024 vintages, 62 million loan-months) with FEMA disaster declarations and Census geography to follow physical climate risk through the whole credit chain: delinquency, pricing, probability of default (PD), loss given default (LGD) and expected loss. A stacked event study shows that hurricanes raise 60+ day delinquency by 1.15 percentage points (about $+42\%$) in the six months after the event, an effect that fades within a year. These delinquencies rarely become losses: disaster-flagged defaults are resolved through forbearance and deferrals, with a measured LGD of 0.04% against 2.8% for other defaults of the same era. Loans near the hurricane coast default about 26% more often than comparable inland loans and pay a rate premium of about 2 bp. In a pre-registered out-of-time test, static climate features add no discriminatory power to a PD scorecard. Finally, an expected-loss stress test of the $35 bn 2021–2024 book shows that a 2008-type recession raises 24-month expected loss to 0.59% of balance, while a severe, simultaneous hurricane hit on every coastal area adds at most 0.01%.**

***Index Terms*—climate risk, mortgage credit, event study, probability of default, loss given default, stress testing, scorecard, LightGBM**

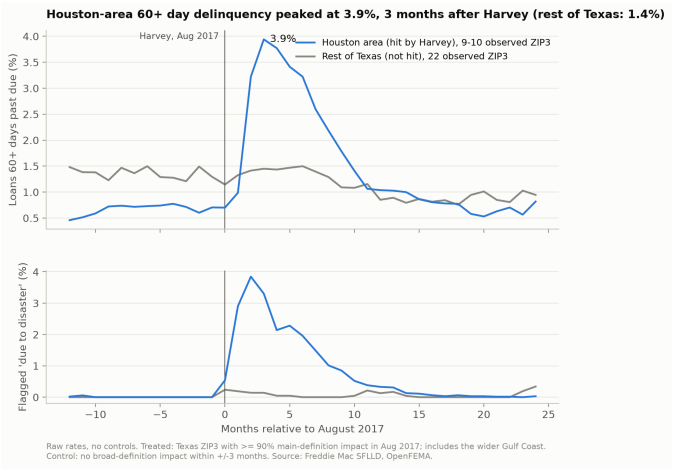


Fig. 1: Houston-area 60+ day delinquency around Hurricane Harvey (August 2017) against the rest of Texas (top), and the share of loans flagged as delinquent due to a disaster (bottom). Raw rates, no controls.

I. Introduction

PHYSICAL climate risk has become a standard item of bank stress testing, but its size in a mortgage portfolio remains an empirical question. A disaster can damage both the collateral and the income of a household, yet the lender's loss depends on what happens *after* a missed payment: whether the loan cures, is modified or ends in foreclosure, and at what price the house is sold.

Hurricane Harvey illustrates the tension (Fig. 1). In the Houston area the share of loans 60+ days past due rose from 0.7% to 3.9% within three months of landfall, while the rest of Texas stayed near 1.4%; almost all of the increase was flagged by servicers as disaster-related, and it was gone a year later. Whether such episodes translate into credit losses, and whether lenders charge for them, is what this work measures.

We follow climate risk through the credit chain on public data and ask four questions: do disasters cause delinquency, and for how long; do coastal loans default more, and are they charged for it; do climate features improve an origination PD model; and what is the expected loss of a recent book under a hurricane and a recession. The contributions are:

- a stacked event study of disasters on delinquency at the ZIP3$\times$month level, with clean controls and a follow-up of what happens to the resulting defaults;
- a pricing analysis comparing, under identical controls, the association of climate variables with the rate spread and with default;
- a pre-registered, out-of-time evaluation of climate features in a weight-of-evidence scorecard and a gradient-boosting benchmark, with through-the-cycle (TTC) calibration;
- an LGD measurement from realised losses with explicit bounds for unresolved defaults, and an expected-loss stress test with a hurricane break-even analysis.

The remainder is organised as follows: Section II covers background and related work; Section III describes the data; Section IV presents the methodology; Section V details practical challenges; Section VI reports results; and Section VII discusses findings and limitations.

TABLE I: Samples used in this work.

Sample	Loans	Default (24 m)
All loans, 2004–2024	1,050,000	1.51%
PD sample	733,940	
Train (2004–2014)	474,368	1.47%
Validation (2015–2018)	144,750	0.60%
Test (2021–2024)	114,822	1.78%
Pricing (30-year, non-HARP)	753,660	

II. Background and Related Work

A. Disasters and Mortgage Credit

Ouazad and Kahn [1] show that after billion-dollar disasters lenders securitise more of their conforming originations, transferring risk to the government-sponsored enterprises. Kousky, Palim and Pan [2] study Hurricane Harvey and find that delinquency rose sharply after the storm, with the most persistent problems for flooded homes outside the designated flood zones. This work extends that line to the full 2004–2024 history of a national sample and to the remaining links of the chain: pricing, PD, LGD and expected loss.

B. Staggered Event Studies

When treatment arrives at different times, two-way fixed-effects regressions can compare treated units with already-treated ones and produce biased averages. Stacked designs, used by Cengiz *et al.* [3], build one sub-experiment per event date with clean controls that are not treated in a surrounding window, and estimate all stacks jointly with stack-specific fixed effects. We follow this approach.

C. Credit Risk Modelling

The weight-of-evidence (WoE) logistic scorecard [4] remains the industry standard for retail PD because it is monotone, interpretable and stable. Gradient-boosted trees such as LightGBM [5] often rank risk better but are harder to validate. Under the Basel framework [6], expected loss is the product of PD, LGD and exposure at default (EAD), and LGD must reflect downturn conditions.

III. Datasets

A. Loans and Default Target

The Freddie Mac Single-Family Loan-Level Dataset [7] sample contains 50,000 loans per origination vintage from 2004 to 2024 (1,050,000 loans) with monthly performance up to March 2026 (62 million loan-months). A loan defaults if, within 24 months of its first payment, it becomes 90+ days delinquent, is acquired as real-estate owned (REO), or exits with a loss. Payoffs are non-defaults; transfers and windows extending past the data cutoff are censored. The PD sample excludes HARP refinancings, date anomalies, windows overlapping the COVID period (March 2020–December 2021) and unknown ZIP3, leaving 733,940 loans (Table I).

B. Disasters

FEMA major-disaster declarations [8] (hurricanes, tropical and coastal storms, floods, dam or levee breaks, fires) with household assistance, excluding statewide and COVID declarations. Counties are mapped to the 890 ZIP3 areas used by Freddie Mac with 2010 Census housing-unit weights [9]. The ex-ante exposure of a ZIP3 is the expected number of disasters per home over the previous ten years, with a three-month availability lag.

C. Rates and Coastline

The Freddie Mac PMMS 30-year rate, obtained from FRED [10], gives the market rate of each origination month. Census 2020 county boundaries and the TIGER coastline give the distance of each county to the Atlantic and Gulf coasts (EPSG:5070 projection); a ZIP3's coastal share is the share of its homes within 50 km.

IV. Methodology

Fig. 2 summarises the pipeline. Every modelling choice is logged with its reason in a decision record (D001–D026), and the full chain is regenerated by one command.

A. Event Study

Each stack is an event month. Treated ZIP3 have at least half of their homes in counties hit by a qualifying disaster; controls are ZIP3 of the same stack with no event in a surrounding window. The outcome is the share of loans 60+ days delinquent, regressed on event-time dummies with ZIP3×stack and Census-division×month×stack fixed effects, weighted by loans and clustered by ZIP3. Parallel trends are tested on the mean and the slope of the lead coefficients: a Monte Carlo exercise showed that a joint Wald test over all leads over-rejects in this setting. The design has 43 stacks, 338 treated ZIP3-events (274 distinct ZIP3) and a median of 68 controls per stack.

B. Pricing

For 30-year non-HARP loans, the spread is the note rate minus the PMMS rate of the origination month (first payment date minus two months). Spread and 24-month default are regressed on the same right-hand side:

$$y_i = \beta_1 \exp_{z(i),t} + \beta_2 \,\mathrm{coast}_{z(i)} + X_i^{\top}\gamma + \alpha_{s(i)\times t} + \varepsilon_i, \quad (1)$$

where exp is the past disaster exposure of the ZIP3, coast its coastal share, X_i contains FICO, LTV and DTI bins, loan size, purpose, occupancy, property type, channel, borrowers, first-time buyer and mortgage insurance, and $\alpha_{s\times t}$ are state×origination-month fixed effects. Standard errors are clustered by ZIP3. The coefficients are conditional associations, not causal effects.

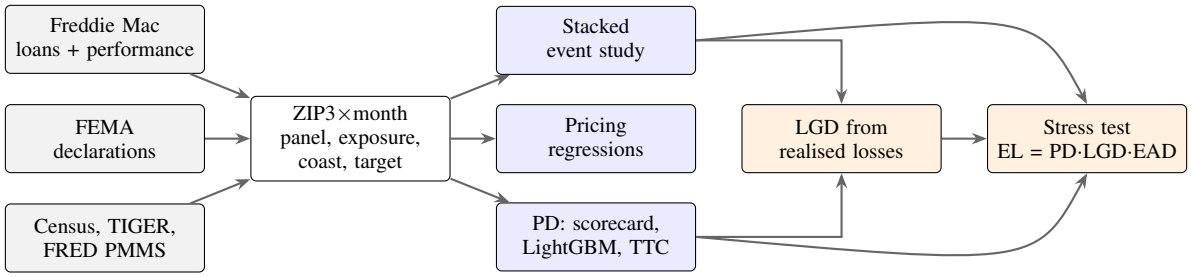


Fig. 2: Pipeline. Public sources are joined into a ZIP3×month panel and loan-level features; the event study, pricing and PD models feed the LGD measurement and the stress test (hurricane effect from the event study, TTC PD from the PD model).

C. PD Model

The split is out of time by vintage (Table I): training includes the 2007–2008 downturn, validation drives every choice, and the test vintages (post-COVID, rising rates) were opened once. Features are known at origination (borrower and loan characteristics, spread over PMMS, state). The main model is a WoE scorecard: quantile bins with separate bins for missing values and for a point mass at the minimum, features selected on training data by information value (IV ≥ 0.02), WoE correlation (≤ 0.8) and coefficient sign. The challenger is LightGBM [5] with early stopping on validation AUC. Calibration is assessed by a level shift a and a slope b on the logit scale; the final PD is anchored through the cycle to the simple average of yearly default rates. Before opening the test, the final models were fixed and committed (pre-registration, D021).

D. LGD and Stress Test

For each default with five years of follow-up, the exposure is the balance at default and the loss is Freddie Mac's reported actual loss (balance + accrued interest + expenses − sale proceeds − recoveries) for loss exits, zero for payoffs and for loans current at the end of the follow-up. Unresolved defaults (transfers, still delinquent, incomplete histories) are bounded between no loss and a loss at the period's severity. The stress test applies

$$\mathrm{EL} = \sum_i \mathrm{PD}_i \times \mathrm{LGD} \times \mathrm{EAD}_i \qquad (2)$$

over 24 months to the 114,822 test-vintage loans ($35.1 bn, EAD = original balance) under three scenarios: *baseline* (TTC PD, LGD of 2017–2021 defaults); *recession* (PD×2.42, the observed-over-TTC ratio of the 2007–2008 vintages, with the 2005–2012 downturn LGD); and *hurricane*, an extra default probability equal to the hurricane effect (1.146 pp) times the coastal share, with every coastal ZIP3 hit at once and every extra delinquency counted as a default, at the LGD of disaster-flagged defaults.

V. Practical Challenges

A. Censoring and Selection

A naive 24-month target would treat recent vintages, whose windows are cut by the data cutoff, as non-defaults, and enrich them in early defaults. The cutoff rule is therefore applied first to every loan, and windows overlapping COVID forbearance are excluded rather than relabelled.

B. Information at Origination

Vintage, servicer and every performance-file field are excluded from the PD features, and the disaster exposure uses a three-month availability lag. Exposure still relies on current FEMA records and static 2010 weights, so it avoids look-ahead by construction of the windows but is not a reconstruction of what lenders knew.

C. Late Flags and Masked Fields

Freddie Mac starts flagging disaster-related delinquency only in 2014, so the cure and LGD comparisons by flag are restricted to later defaults. Losses are published for most but not all loss exits; missing amounts are imputed at the period's severity, and the LGD bounds make the assumption explicit.

D. Zero-Inflated Binning

Recent-disaster counts are zero for most loans. Plain quantile edges collapsed into a single bin, which made the population stability index (PSI) of the feature exactly zero and hid a real drift. A dedicated bin for the point mass at the minimum fixed both the WoE and the PSI; all PD results were rerun after the fix.

VI. Results

A. Disasters Cause Short-Lived Delinquency

Across all events, delinquency rises by 0.46 pp over months 1–6 (s.e. 0.09; baseline 2.19%), with no detectable pre-trend (mean test $p = 0.41$, slope test $p = 0.30$). The effect is driven by hurricanes (Fig. 3): $+1.15$ pp (s.e. 0.15, 173 events) on a baseline of 2.70%, i.e. about $+42\%$, peaking two to three months after the event and fading within a year. Fires show a smaller effect ($+0.24$ pp) with a lead profile that calls for caution. Flood estimates fail the pre-trend test ($p < 0.01$) and are not interpreted causally. The effect grows with treatment intensity (0.46, 0.50 and 0.58 pp for hit-share thresholds of 0.5, 0.75 and 0.9).

Table II follows defaults from 2014 onwards for 24 months. Disaster-related defaults are real payment interruptions but almost never end in foreclosure: the shock is to liquidity rather than solvency, and servicers bridge it.

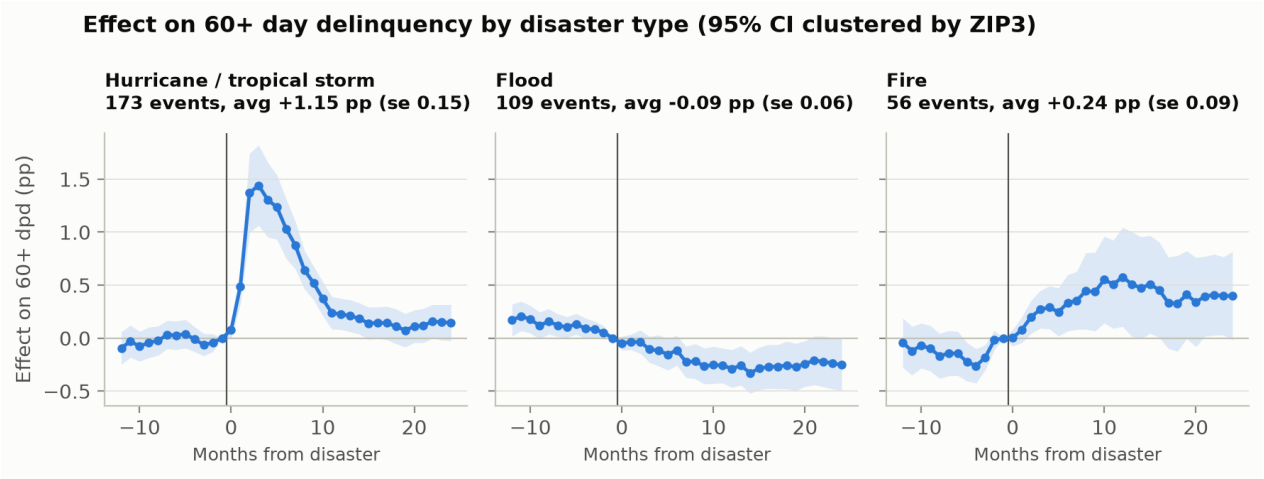


Fig. 3: Effect of disasters on 60+ day delinquency by disaster type, relative to the month before the event (95% confidence intervals clustered by ZIP3).

TABLE II: Outcome of defaults over 24 months, by flag at default (defaults from 2014).

Flag at default	N	Loss/REO	Current +24 m
Disaster	796	1.0%	63.7%
Forbearance only	312	2.6%	49.7%
Neither	2,036	16.0%	32.2%

TABLE III: Pre-registered test (2021–2024 vintages, 114,822 loans, 2,049 defaults); 95% bootstrap intervals over loans.

Model	AUC	Gini
Scorecard + coast (primary)	0.782 [0.774, 0.792]	0.564
Scorecard (base)	0.782 [0.774, 0.792]	0.565
LightGBM (benchmark)	0.797 [0.789, 0.806]	0.595
Primary − base	−0.0002 [−0.0008, +0.0003]	
LightGBM − base	+0.015 [+0.011, +0.020]	

B. Coastal Risk Is Barely Priced

Past disaster exposure is associated with neither the spread nor default (Fig. 4). A fully coastal ZIP3 is associated with a spread higher by 2.0 bp [1.0, 3.0] and a 24-month default probability higher by 0.42 pp [0.21, 0.64] on a mean of 1.60%, i.e. about +26%. The relative effect is similar before 2012 (+0.58 pp on 2.25%) and after (+0.26 pp on 0.98%), so it is not an artefact of the 2007–2008 bust. Spread over 24 months, the extra default probability is about 0.21 pp per year; with the LGD of Section VI-D it implies an extra expected loss of roughly 0.2 bp per year in benign conditions and 6 bp in a downturn. The 2 bp premium sits between the two: coastal risk looks priced on average over a cycle but not for a severe downturn.

C. Climate Features Do Not Improve PD

On validation the scorecard reached an AUC of 0.814 and LightGBM 0.816, but the scorecard over-predicted defaults by a factor of about two, uniformly across deciles ($a = -0.77$, $b = 0.91$): it ranks loans well, but its level reflects the crisis-heavy training period. Without macroeconomic variables the model cannot forecast the cycle, so its PD is anchored through the cycle to 1.105%. Fig. 5 shows that the TTC PD tracks underwriting quality (2.3% for 2007 originations) but not the macro shock (4.9% observed).

On validation, static coast features raised the scorecard AUC by 0.0022 [0.0015, 0.0028], while time-varying exposure features drifted (PSI 0.15–0.21) and degraded LightGBM. The pre-registered test (Table III) did not replicate two validation conclusions, and both are reported. First, the climate gain vanishes. Second, LightGBM beats the scorecard out of time by 0.015 AUC, plausibly through risk-layering interactions that an additive scorecard cannot represent; the pre-registered choice of the scorecard is kept but flagged as a model-risk finding. The TTC PD under-predicts the test vintages (0.94% against 1.78% observed). With a pre-registered credit-default target that ignores defaults cured within twelve months (D022), the under-prediction falls from ×1.95 to ×1.41, climate still adds nothing (+0.0002) and the LightGBM advantage grows to +0.030 [0.018, 0.041]; after 2015, 52–71% of cures go through a modification or deferral.

D. LGD Moves with the Cycle

Table IV shows that in a downturn there are more defaults *and* each costs more, as house prices fall. Disaster-related defaults lose less than other defaults under every assumption; with about 180 resolved disaster defaults and a single loss exit the central value is fragile, but even the high bound stays near 3%.

E. The Cycle Dominates Expected Loss

A 2008-type recession raises the 24-month expected loss of the 2021–2024 book from 0.009% to 0.592% of balance, about 67 times the baseline (Fig. 6, Table V). The hurricane adds 0.0001% of balance, 0.010% at the high disaster LGD. For the hurricane to cost as much as the recession, its default

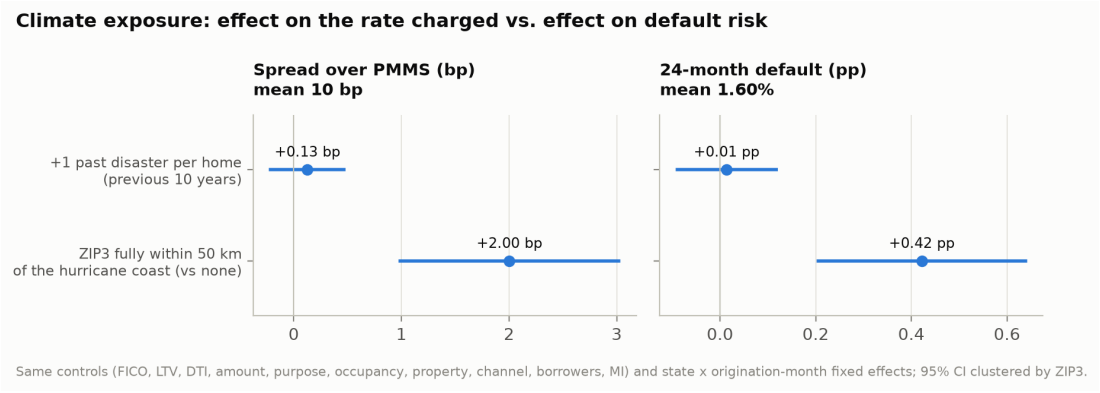


Fig. 4: Climate variables: association with the spread over PMMS and with 24-month default under the same controls and fixed effects (PD sample, 95% CI).

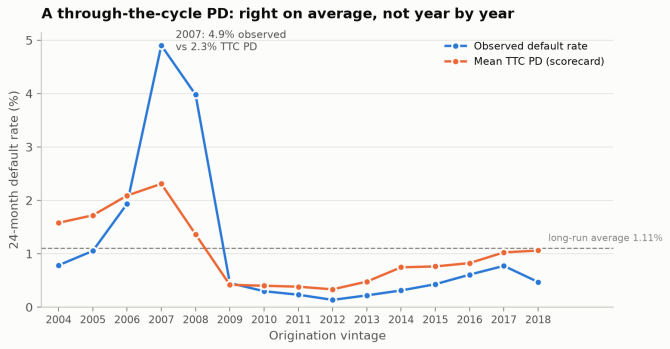


Fig. 5: TTC PD of the scorecard and observed default rate by origination vintage (in-sample, 2004–2018).

TABLE IV: LGD (% of exposure at default): central estimate over resolved defaults, with bounds for unresolved defaults.

Defaults	Central	Low	High
2005–2008	31.5	23.0	35.9
2009–2012	26.3	20.2	31.5
2013–2016	4.6	3.7	7.6
2017–2021	1.0	0.8	3.6
From 2014: disaster	0.04	0.03	3.08
From 2014: forbearance only	0.50	0.42	2.71
From 2014: neither	2.77	2.11	5.44

effect would have to be about 4,000 times larger at the central LGD (impossible, as it would exceed a 100% probability) or 57 times larger at the high LGD, i.e. about 65% of coastal loans defaulting because of the storm. Because the TTC PD under-predicts recent vintages, the baseline is also computed with PD×1.90 (0.017%); the hurricane increment does not depend on the PD level.

VII. Discussion and Conclusion

A. Discussion

Climate risk is visible in mortgage data, but mostly as a liquidity shock rather than a credit loss. Hurricanes cause a sharp, short-lived rise in delinquency that servicers absorb through

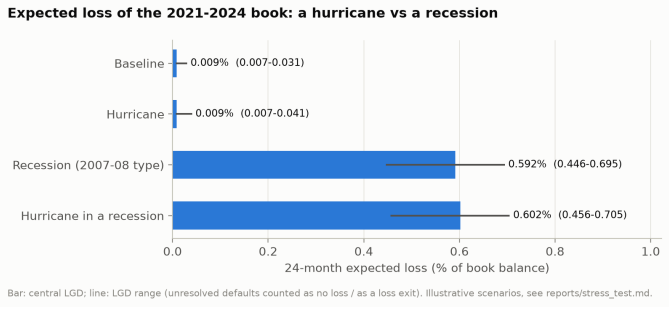


Fig. 6: Expected loss of the 2021–2024 book by scenario (% of balance). Bars: central LGD; lines: LGD range.

TABLE V: 24-month expected loss of the 2021–2024 book ($35.1 bn).

Scenario	EL (%)	Range (%)	$ m
Baseline	0.009	0.007–0.031	3.1
Hurricane (increment)	0.0001	0.0001–0.010	0.05
Recession (2007–08)	0.592	0.446–0.695	207.4
Hurricane in recession	0.602	0.456–0.705	211.0

forbearance and deferrals, so disaster-related defaults almost never end in foreclosure. Coastal loans carry a structurally higher default rate and a small rate premium, but once standard underwriting variables and the state are known, static climate exposure adds nothing to default prediction. In expected-loss terms the cycle dominates: a recession costs the 2021–2024 book dozens of times more than an extreme hurricane season.

B. Limitations

The results cover the lender's or guarantor's credit loss over 24 months under current servicing practices; property damage, insurance costs, uninsured flood losses and long-run coastal house-price risk are outside this measure, and so is the cost of workouts borne by the guarantor. The PD model has no macroeconomic variables, so its TTC PD is not a forecast of current conditions. Causal claims are limited to hurricanes; flood effects are not identified, and the pricing and PD results are conditional associations. Bootstrap intervals

resample loans rather than ZIP3 and are therefore too narrow; the disaster LGD rests on few loss events; and the hurricane scenario is an illustrative, deliberately severe mapping of a delinquency effect into defaults.

C. Conclusion

Following climate risk through the whole credit chain changes the answer to the question in the title. Hurricanes clearly cause delinquency, coastal loans clearly default more, and the rate premium is small; yet because disaster-related defaults cure, the expected credit loss they create is negligible next to an ordinary recession. The open questions are those this measure cannot see: the insurance market, the cost of servicer workouts and the long-run repricing of coastal housing.

Reproducibility

The code, configuration and decision log are public in the project repository. `make all` downloads the public data (Freddie Mac files require a free registration), builds every dataset and regenerates every report and figure; `make check` runs the linter and 91 tests, including synthetic-data tests in which known effects must be recovered. Each modelling choice is recorded with its reason in `DECISIONS.md` (D001–D026), including the pre-registration of the final test (D021–D022) and the fixes made after an external review (D026).

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